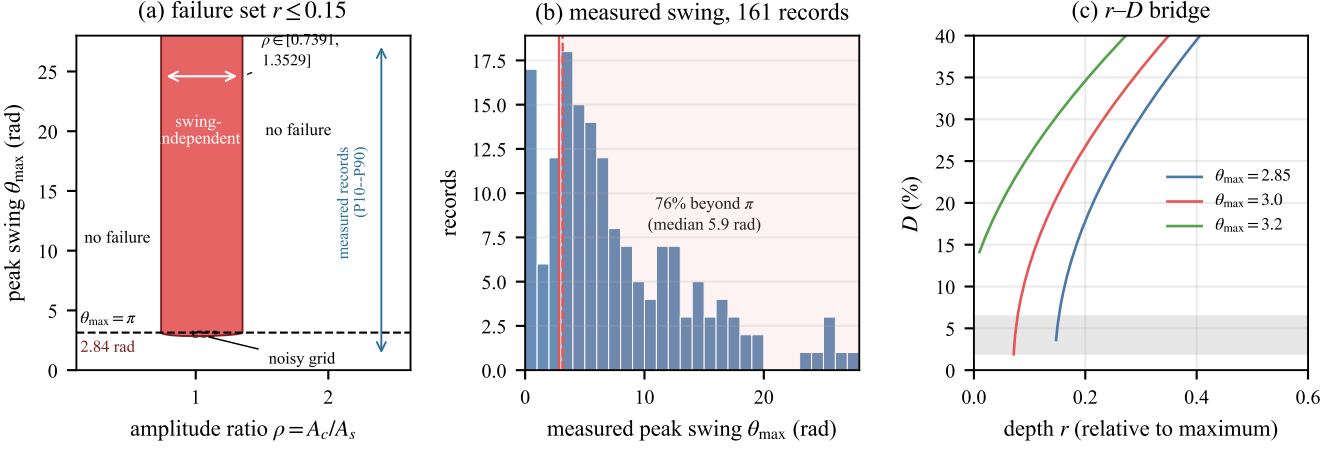
(a) failure set $r \le 0.15$
$\rho \in [0.7391, 1.3529]$
swing-independent
no failure
no failure
measured records (P10–P90)
$\theta_{max} = \pi$
2.84 rad
noisy grid
peak swing θ_{max} (rad)
amplitude ratio $\rho = A_c/A_s$
(b) measured swing, 161 records
76% beyond π
(median 5.9 rad)
records
measured peak swing θ_{max} (rad)
(c) r–D bridge
$\theta_{max} = 2.85$
$\theta_{max} = 3.0$
$\theta_{max} = 3.2$
D (%)
depth r (relative to maximum)